

2.3 LSTMs and GRUs

Gates keep a near-1 path through time. LSTM has a cell; GRU does not.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Gates are learned valves. They let a recurrent cell **keep** a bit of memory for many steps instead of overwriting it every time. LSTM and GRU are the two you will meet. They do not remove vanishing gradients; they make a near-1 path available.

1. LSTM: a highway plus three gates

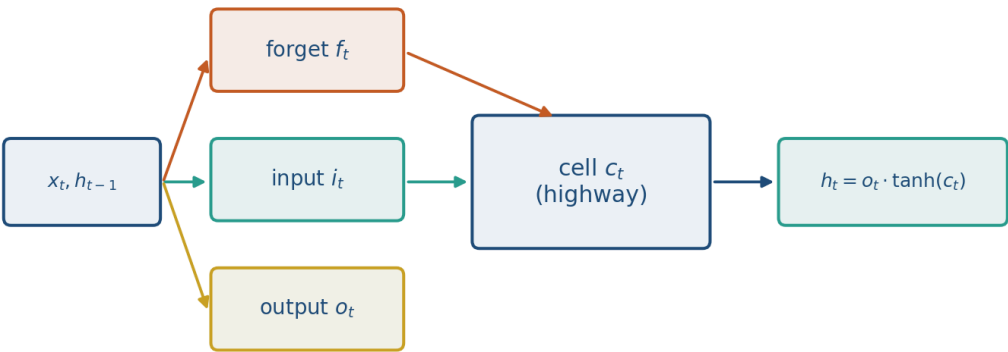
Hochreiter and Schmidhuber (1997). A **cell state** c_t moves down the chain with mostly multiplies and adds, not a stack of tanhs. Three sigmoid gates, each in $(0, 1)$:

Gate	Role
Forget f_t	How much of c_{t-1} to keep
Input i_t	How much new candidate to write
Output o_t	How much of $\tanh(c_t)$ becomes h_t

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t).$$

If $f_t \approx 1$ and $i_t \approx 0$, the cell copies. That is the long-range path.

LSTM: a cell highway plus learned gates



The candidate $\tilde{c}_t$ is a tanh of a linear mix of x_t and h_{t-1} . You do not need every matrix name on the first pass. You need: **forget** × **old**, **plus input** × **new**.

If $f_t = 1$ for many steps, $\partial c_t / \partial c_{t-1}$ can stay near 1 along that coordinate. That is why the vanishing plot in 2.2 had a gated curve.

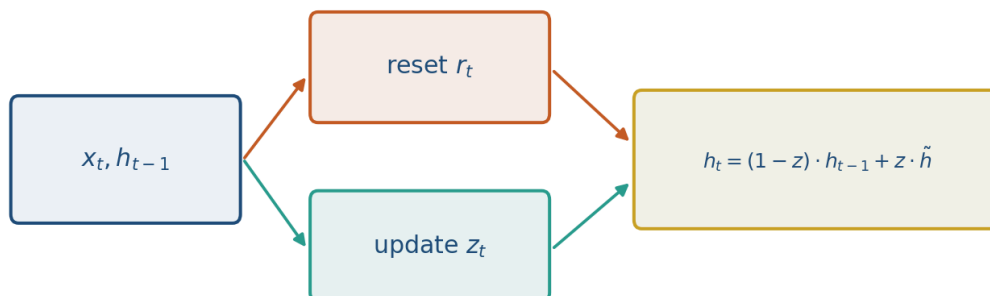
2. GRU: two gates, no extra cell

Cho et al. (2014). Reset r_t and update z_t . The hidden state *is* the memory.

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t.$$

Fewer parameters, often similar quality on medium sequences. Use LSTM when you want an explicit cell; use GRU when you want a smaller default.

GRU: two gates, no separate cell state



Read z_t as “how much to replace.” If $z_t \approx 0$, you copy h_{t-1} . Reset r_t controls how much past goes into the candidate $\tilde{h}_t$.

3. What gates do not buy you

The loop is still sequential. You still cannot fill a GPU the way a transformer can. Long context is better than vanilla, worse than attention. For this course, gates are the reason we can talk about “memory” before Week 3, and a baseline in speech/time-series projects.

Bidirectional LSTM: run one net forward and one backward, concatenate. Fine for classification. Illegal for next-token generation (you would peek at the future).

4. Teaching this note

30–40 minutes. Table of three LSTM gates, then the copy regime $f \approx 1, i \approx 0$, then the numeric cell update. GRU as “two knobs, hidden state is the memory.” Play StatQuest LSTM **0:00–12:00** (forget/input intuition). Assign the GRU video as a 10-minute clip after class, or play **0:00–8:00** if the room is still with you.

5. Worked example

Scalar LSTM cell (drop the output gate for a minute). Suppose $c_{t-1} = 2, f_t = 0.9, i_t = 0.1, \tilde{c}_t = 5$:

$$c_t = 0.9 \cdot 2 + 0.1 \cdot 5 = 1.8 + 0.5 = 2.3.$$

Most of the old memory survived; a little new content landed. If instead $f_t = 0$, $c_t = 0.5$ and the 2 is gone.

Copy regime: $f_t = 1$, $i_t = 0 \Rightarrow c_t = c_{t-1}$. Ten such steps: $c_{t+10} = c_t$. Gradient along that path is $1^{10} = 1$.

GRU: $h_{t-1} = 4$, $z_t = 0.25$, $\tilde{h}_t = 0$ gives $h_t = (1 - 0.25) \cdot 4 + 0.25 \cdot 0 = 3$. You kept 75% of the past.

6. Where students get stuck

- Mixing up forget vs. input (zeroing f when they meant “write nothing new”).
 - Claiming LSTMs “solved vanishing gradients” in every coordinate, always.
 - Using a bidirectional LSTM for a generator that must not see $t + 1$.
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7. Video

Watch [StatQuest: Long Short-Term Memory \(LSTM\), Clearly Explained](#).

Pause when the forget gate is a number in $(0, 1)$ multiplying the cell, and when the cell is drawn as a highway. Then [StatQuest: Gated Recurrent Units \(GRU\), Clearly Explained](#) — play the update-gate mix; skip if time is gone.

8. Practice

1. If the forget gate is stuck at 0, what happens to c_t ?
2. Why is a bidirectional LSTM the wrong cell for a language model that must emit token $t + 1$?
3. Name one reason you might still pick a GRU over a transformer in a project.
4. Compute c_t for $c_{t-1} = 1$, $f_t = 0.5$, $i_t = 0.5$, $\tilde{c}_t = -1$. Then compute h_t if $o_t = 1$ and $h_t = o_t \tanh(c_t)$.
5. GRU with $h_{t-1} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $z_t = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\tilde{h}_t = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$. Compute h_t . Which coordinate copied, and which replaced?